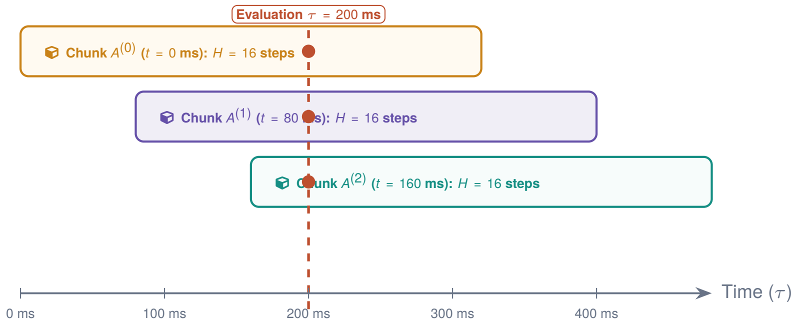
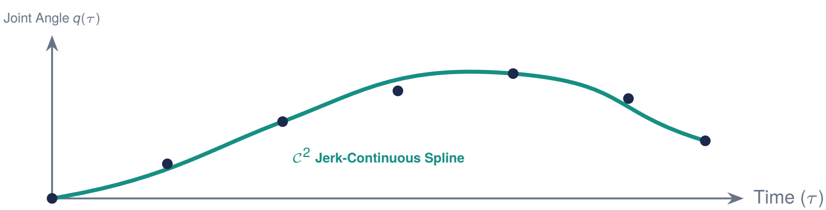
RECEDING-HORIZON ACTION CHUNKING & TEMPORAL ENSEMBLING

Horizon $H = 16$, Stride $K = 4$, $\lambda = 0.15$

1. Overlapping Action Chunks at Successive Cycles ($A^{(0)}, A^{(1)}, A^{(2)}$)



2. Continuous Synthesized Trajectory $\mathbf{a}^*(\tau)$ Executed by MCU at 1000 Hz



Exponential Temporal Ensembling

At physical time τ, multiple overlapping chunks $A^{(i)}$ provide candidate predictions $\mathbf{a}^{(i)}_{\tau - t_i}$. The blended command $\mathbf{a}^*(\tau)$ is:

$$\mathbf{a}^*(\tau) = \frac{\sum_{i=0}^{N-1} w_i \cdot \mathbf{a}^{(i)}_{\tau - t_i}}{\sum_{i=0}^{N-1} w_i}$$

where weights decay exponentially with the age of the prediction:

$$w_i = \exp\left(-\lambda(\tau - t_i)\right)$$

Physical AI Systems Impact:

- **Noise Filtering:** Suppresses high-frequency diffusion sampling jitter.
- **Delay Amortization:** 50 ms inference latency is amortized across 320 ms of motion.
- **Continuity Guarantee:** Prevents discrete velocity jumps at chunk borders.

Systems Performance Law: Action chunking with temporal ensembling converts stochastic, discrete neural inference into a continuous reference reservoir. By executing $K = 4$ steps before blending the next chunk, the real-time MCU never starves, maintaining 1000 Hz motor control with zero mechanical shock.